Advanced Programming Practice · 21CSC203P

Unit 1 · Session 10 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 10 : SLO – 1 | Logic Programming Simulation

Activity (Solutions)

• 1) Expert-system rule in Java: int age = 20; if (age > 18) System.out.println("Eligible to vote"); else System.out.println("Not eligible"); — mimics the Prolog rule eligible(X) :- age(X,A), A > 18.

• 2) Decision rules → conditions: rule 'discount if member AND purchase > 1000' → boolean discount = isMember && amount > 1000; rule 'alarm if temp > 40 or smoke detected' → if (temp > 40 || smoke) alarm();

• 3) Rule-based quiz app design: class Quiz { String question; Map<Character,String> options; char key; boolean check(char ans){ return ans == key; } } — each question object stores its rule (correct key); the engine asks, reads input, applies the rule check(), and scores: score += q.check(ans) ? 1 : 0.